Review

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Question

Prove that

$$PSUBSET_{DFA} = \{\langle A, B\rangle | A \text{ and } B \text{ are DFAs and } L(A) \subset L(B)\}$$

is decidable.

- Recall that you need to perform two tasks:
  1. Construct a TM that you think it decides $PSUBSET_{DFA}$
  2. Prove that your TM is actually a decider for $PSUBSET_{DFA}$
- Recall $L(A) \subset L(B)$ (proper subset)
  - $L(A) \neq L(B)$ and $L(A)$ is a subset of $L(B)$

$PSUBSET_{DFA}$ is Decidable

Solution: Construct a TM M' to decide $PSUBSET_{DFA}$ as follows:

$M' =$ "On input $\langle A, B \rangle$ where A and B are DFAs:

1. Run TM F on input $\langle A, B \rangle$
2. If F accepts $\langle A, B \rangle$, *reject*.
3. Construct a DFA C such that $L(C) = L(A) \cap \overline{L(B)}$
4. Run TM T on input $\langle C \rangle$
5. If T accepts $\langle C \rangle$, *accept*. If T rejects $\langle C \rangle$, *reject*."

Next, we need to prove that the above TM M' is a decider for $PSUBSET_{DFA}$.

$PSUBSET_{TM}$ is Decidable

To show that TM M' is a decider for $PSUBSET_{DFA}$, we need to perform the following:

- Show that if $\langle A, B \rangle \in PSUBSET_{DFA}$, M' will accept $\langle A, B \rangle$.

  Assume that $\langle A, B \rangle \in PSUBSET_{DFA}$. Since $\langle A, B \rangle \in PSUBSET_{DFA}$, A and B are DFAs and $L(A)$ is a proper subset of $L(B)$. In other words, $L(A)$ is not equal to $L(B)$ and $L(A)$ is a subset of $L(B)$. Since $L(A)$ is not equal to $L(B)$, $\langle A, B \rangle \notin EQ_{DFA}$. Since F is a decider for EQ_{DFA}, by running F on input $\langle A, B \rangle$, F will reject $\langle A, B \rangle$. Since $L(A)$ is a subset of $L(B)$, $L(C)$ will be empty. Since $L(C)$ is empty, $\langle C \rangle \in E_{DFA}$. Since T is a decider for E_{DFA}, by running T on input $\langle C \rangle$, T will accept $\langle C \rangle$. Since T accepts $\langle C \rangle$, M' accepts $\langle A, B \rangle$.
- Show that if $\langle A, B \rangle \notin \text{PSUBSET}_{TM}$, M' will reject $\langle A, B \rangle$.

  Assume that $\langle A, B \rangle \notin PSUBSET_{DFA}$. Since $\langle A, B \rangle \notin PSUBSET_{DFA}$, A and B are DFAs and $L(A)$ is not a proper subset of $L(B)$. In other words, $L(A) = L(B)$ or $L(A)$ is not a subset of $L(B)$.

  Suppose $L(A) = L(B)$. Since $L(A) = L(B)$, $\langle A, B \rangle \in EQ_{DFA}$. Since F is a decider for EQ_{DFA}, by running F on input $\langle A, B \rangle$, F will accept $\langle A, B \rangle$. Since F accepts $\langle A, B \rangle$, M' rejects $\langle A, B \rangle$.

  Suppose $L(A)$ is not a subset of $L(B)$. Since $L(A)$ is not a subset of $L(B)$, $L(C)$ will not be empty. Since $L(C)$ is not empty, $\langle C \rangle \notin E_{DFA}$. Since T is a decider for E_{DFA}, by running T on input $\langle C \rangle$, T will reject $\langle C \rangle$. Since T rejects $\langle C \rangle$, M' rejects $\langle A, B \rangle$.

This shows that M' is a decider for $PSUBSET_{DFA}$ DFA. Therefore, $PSUBSET_{DFA}$ is decidable.

Question

Prove that

$$ALLBUTONE_{\text{TM}} = \{\langle M \rangle | M \text{ is a TM that accepts all but one string}\}$$

is undecidable. Note that if TM M accepts all but one string, $L(M) = \Sigma^* - \{w\}$ for a string w.

- Recall that there are two methods:
  1. Show that if $ALLBUTONE_{\text{TM}}$ is decidable, A_{TM} is decidable
  2. Use mapping reducibility to show that $A_{\text{TM}} \leq_m ALLBUTONE_{\text{TM}}$

$ALLBUTONE_{\text{TM}}$ is Undecidable (Method 1)

Assume that $ALLBUTONE_{\text{TM}}$ is decidable. Since $ALLBUTONE_{\text{TM}}$ is decidable, there exists a TM R that decides $ALLBUTONE_{\text{TM}}$. Construct a TM S to decide A_{TM} as follows:

$S =$"On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct a TM M' as follows:

   $M' =$"On input x:

   1. If $x == \varepsilon$, *reject*.
   2. Run M on input w.
   3. If M accepts w, *accept*. If M rejects w, *reject*."
2. Run TM R on input $\langle M' \rangle$.
3. If R accepts $\langle M' \rangle$, *accept*. If R rejects $\langle M' \rangle$, *reject*."

Next, we need to show that the above TM S is a decider for A_{TM}.

$ALLBUTONE_{TM}$ is Undecidable (Method 1)

To show that TM M' is a decider for $ALLBUTONE_{TM}$, we need to perform the following:

- If $\langle M, w \rangle \in A_{TM}$, S will accept $\langle M, w \rangle$.

  Assume that $\langle M, w \rangle \in A_{TM}$. Since $\langle M, w \rangle \in A_{TM}$, M is a TM that accepts w. Since M accepts w, from the description of TM M', $L(M') = \Sigma^* - \{\varepsilon\}$. Since $L(M') = \Sigma^* - \{\varepsilon\}$, $\langle M' \rangle \in ALLBUTONE_{TM}$. Since R is a decider for $ALLBUTONE_{TM}$, by running R on input $ALLBUTONE_{TM}$, R will accept $\langle M' \rangle$. Since R accepts $\langle M' \rangle$, S accepts $\langle M, w \rangle$.

- If $\langle M, w \rangle \notin A_{TM}$, S will reject $\langle M, w \rangle$.

  Assume that $\langle M, w \rangle \notin A_{TM}$. Since $\langle M, w \rangle \notin A_{TM}$, M is a TM that does not accept w. Since M does not accept w, from the description of TM M', $L(M') = \emptyset$. Since $L(M') = \emptyset$, $L(M') \neq \Sigma^* - \{w\}$ for a string w. In other words, $\langle M' \rangle \notin ALLBUTONE_{TM}$. Since R is a decider for $ALLBUTONE_{TM}$, by running R on input $ALLBUTONE_{TM}$, R will reject $\langle M' \rangle$. Since R rejects $\langle M' \rangle$, S rejects $\langle M, w \rangle$.

This shows that S is a decider for A_{TM}. In other words A_{TM} is decidable. This contradicts with the fact that A_{TM} is undecidable. Therefore, $ALLBUTONE_{TM}$ is undecidable.

$ALLBUTONE_{\text{TM}}$ is Undecidable (Method 2)

Need to show that $A_{\text{TM}} \leq_m ALLBUTONE_{\text{TM}}$. Construct a TM F as follows:

$F =$"On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct a TM M' as follows:

   $M' =$"On input x:

   1. If $x == \varepsilon$, *reject*.
   2. Run M on input w.
   3. If M accepts w, *accept*. If M rejects w, *reject*."

2. Output $\langle M' \rangle$.

Next, we need to show that the above TM F is a reduction from A_{TM} to $ALLBUTONE_{\text{TM}}$. In other word, we have to show that it satisfies:

$$\langle M, w \rangle \in A_{\text{TM}} \leftrightarrow F(\langle M, w \rangle) \in ALLBUTONE_{\text{TM}}$$

$ALLBUTONE_{TM}$ is Undecidable (Method 2)

To show that TM F is a reduction from A_{TM} to $ALLBUTONE_{TM}$, we need to perform the following:

- If $\langle M, w\rangle \in A_{TM}$, $F(\langle M, w\rangle) \in ALLBUTONE_{TM}$.

  Assume that $\langle M, w\rangle \in A_{TM}$. Since $\langle M, w\rangle \in A_{TM}$, M is a TM that accepts w. Since M accepts w, from the description of TM M', $L(M') = \Sigma^* - \{\varepsilon\}$. Since $L(M') = \Sigma^* - \{\varepsilon\}$, $\langle M'\rangle \in ALLBUTONE_{TM}$. Since $F(\langle M, w\rangle) = \langle M'\rangle$, $F(\langle M, w\rangle) \in ALLBUTONE_{TM}$.
- If $\langle M, w\rangle \notin A_{TM}$, $F(\langle M, w\rangle) \notin ALLBUTONE_{TM}$.

  Assume that $\langle M, w\rangle \notin A_{TM}$. Since $\langle M, w\rangle \notin A_{TM}$, M is a TM that does not accept w. Since M does not accept w, from the description of TM M', $L(M') = \emptyset$. Since $L(M') = \emptyset$, $L(M') \neq \Sigma^* - \{w\}$ for a string w. In other words, $\langle M'\rangle \notin ALLBUTONE_{TM}$. Since $F(\langle M, w\rangle) = \langle M'\rangle$, $F(\langle M, w\rangle) \notin ALLBUTONE_{TM}$.

This shows that F is a reduction from A_{TM} to $ALLBUTONE_{TM}$. Since A_{TM} is undecidable, $ALLBUTONE_{TM}$ is undecidable.

Question

Consider the following language

$NOTPRIME = \{\langle n \rangle | n$ is a natural number and n is not a prime number$\}$

Perform the following:

1. Construct a verifier for the language *NOTPRIME*
2. Construct an NTM that decides *NOTPRIME*

A Verifier of *NOTPRIME*

- Given a language A, a verifier V of the language A is a TM satisfying

$$A = \{w \mid V \text{ accepts } \langle w, c \rangle \text{ for some } c\}$$

- From the above description
  - Given a string $w \in A$, V must accept $\langle w, c \rangle$ for some c
  - Given a string $w \notin A$, V must reject $\langle w, c \rangle$ for all c
- A verifier for *NOTPRIME* must satisfies

$$\textit{NOTPRIME} = \{\langle n \rangle \mid V \text{ accepts } \langle n, c \rangle \text{ for some } c\}$$

- An example of a verifier for the language *NOTPRIME*:

  $V =$"On input $\langle n, c \rangle$ where n and c are natural numbers:

  1. If $c == 1$ or $c \geq n$, *reject*.
  2. If $n \mod c == 0$, *accept*."

An NTM that decides *NOTPRIME*

- Think about how brute force algorithm should work
  - Generate all possible strings between 2 and $n-1$ (inclusive)
  - If n is divisible by one of them, n is not a prime number
- An example of an NTM that decides the language *NOTPRIME*:

  $N =$ "On input $\langle n \rangle$ where n is a natural numbers:

  1. Select a number c where c is nondeterministically selected between 2 and $n-1$ (inclusive)
  2. If $n \mod c == 0$, *accept*."